Province of the
EASTERN CAPE
EDUCATION

NATIONAL SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2019

CIVIL TECHNOLOGY: WOODWORKING MARKING GUIDELINE

MARKS: **200**

This marking guideline consists of 11 pages, including 2 answer sheets.

QUESTION 1: SAFETY AND MATERIALS (GENERIC)

1.1 Personal protective equipment (1)

1.2 Any TWO requirements of boots on a building site: (2 x 1) (2)
- Sturdy
- Non-slip
- Metal reinforcements in the toes

1.3 Safety precautions for small plant equipment:

1.3.1 To ensure that the equipment is in a good, working condition. (1)

1.3.2 Less chance of inhaling the hazardous fumes of the engines. (1)

1.3.3 Avoiding any possible injuries. (1)

1.3.4 Insufficient training could lead to injuries and damaged equipment. (1)

1.4 Safe stacking of material:

1.4.1 Ladders or any similar answer (1)

1.4.2 Any TWO factors that should not be affected:
- Ventilation
- Lighting
- Fire-fighting equipment (2 x 1) (2)

1.4.3 3 x 500 mm (1) = 1 500 mm of 1,5 m (2) (2)

1.4.4 Can easily hook onto or bump against protruding parts and that could cause the stack to fall over. (1)

1.5 Cement (1) and fine sand (2) (2)

1.6 Any ONE example of a fine aggregate:
- Sand
- Silt
- Clay (1)

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1.7 Any ONE purpose of lime:
- Increase the plasticity of the mixture
- Makes the mixture more workable
- Deduction of cracks (1)

1.8 Any TWO board products for panelling work:
- Plywood
- Block board
- Hardboard / Masonite(2 x 1) (2)

1.9 Any TWO uses of stainless steel:
- Sinks
- Wash tubs
- Water taps
- Water traps
- Extractor fans
- Any similar answers (2 x 1) (2)

1.10 Iron (1)

1.11 Two or more metals, or metals and non-metals are combined (1), to form a new, permanent metal (2), with enhanced qualities. (3) (3)

1.12 Any TWO uses of safety glass:
- Sliding doors
- Exterior doors with glass panels
- Shower cubicle and doors
- Bath glass screens
- Balustrades of staircases (2 x 1) (2)

1.13 Any ONE use of a mastic sealant:
- Adheres to almost any material (wood, glass, aluminium, concrete etc.)
- For filling cracks and sealing areas exposed to water
- Used in construction projects (roofing and brickwork) (1)

1.14 Can be reshaped (1) when reheated (2). (2)

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QUESTION 2: EQUIPMENT, TOOLS AND GRAPHICS (GENERIC)

2.1 Name the tools in FIGURES 2.1.1 to 2.1.4 and name ONE use of each.

2.1.1 Claw hammer
Any ONE use:
- General carpentry / Driving in nails
- Remove nails (2)

2.1.2 Plastering trowel
- Smooth finishing for walls / Plaster work (2)

2.1.3 Club hammer
Any ONE use:
- Driving brick bolster / cold chisel
- Where heavy hammering is needed
- Driving pegs into the ground (2)

2.1.4 Angle grinder
Any ONE use:
- Cutting stone / concrete / tiles / metals
- Can be used as a grinder (2)

2.2 2.2.1 Bench grinder (1)

2.2.2 Portable circular saw / Radial arm saw (1)

2.3 Identify the tool in FIGURE 2.3 and name TWO uses of it.

Plate compacter

Any TWO uses:
- Compacting disturbed / loose soil up to 150 mm
- Tampering fillings for hard-core layer
- Compacting soil for paving bricks (3 x 1) (3)

2.4
- Wipe clean after use
- Do not allow mortar / concrete / screed to dry on it
- Store in a dry place (Any 2 x 1) (2)

2.5 Similar answer:
The jaws of the universal pliers cannot open big enough (1)

2.6 2.6.1 Section view (1)

2.6.2 2.6.A – Purlin
2.6.B – Beam filling
2.6.C – Rafter
2.6.D – Window sill
2.6.E – Foundation
2.6.F – Wall plate
2.6.G – Wall tie
2.6.H – Lintel
2.6.I – Damp proof coarse
2.6.J – Hard-core (10)

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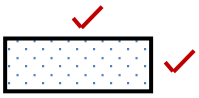
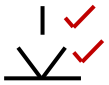
2.6.3 114 x 38 (2)

2.6.4 Bind 2 wall leaves (1)

2.7
- Pitch of roof
- Type of roof covering
- Barge board
- Facia board
- Gutters
- Downpipes (Any 4 x 1) (4)

2.8 2.8.1 Plaster  (2)

2.8.2 Undressed wood (2)

2.8.3 Invert level  (2)

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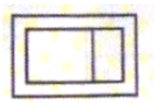
QUESTION 3: QUANTITIES, JOINING AND GRAPHIC (GENERIC)

3.1 Make neat sketches to illustrate the following symbols on a floor plan:

3.1.1 Grease trap  (2)

3.1.2 DPC (Damp-proof course) (2)

3.1.3 Staircase (2)

3.2 **Use ANSWER SHEET 1** (12)

3.3
- Basic sealant against air and leakage
- Enhance materials
- Used in aviation
- Construction repairs (Any 3 x 1) (3)

3.4
- Apply enough adhesive to both sides of the areas to be bonded
- Allow to dry
- After drying, hold the two pieces of material together or clamp, roll or press them together (3)

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3.5
- Joins PVC pipes
- Clear / transparent
- Dries quickly (Any 1 x 1) (1)

3.6
- To allow light into building
- To prevent rain, wind, dust and insects from entering
- Enhance the aesthetic qualities of a structure (3)

3.7
- Polythene: strong and light / becomes brittle when exposed to sunlight / can be used in underground waterproofing / can be reshaped / remoulded after heating.
- Polyvinyl chloride: can be reshaped / solid material / two types available flexible and rigid / good insulating properties / not dissolved by alcohol. (2)

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QUESTION 4: JOINING, WINDOWS, DOORS AND WALL PANELLING (SPECIFIC)

4.1 **ANSWER SHEET 2** (20)

4.2 4.2.1 Prevents rainwater from being blown into the casement and penetrating the room. (1)

4.2.2 To hold or secure the glass / to prevent the glass from falling out of the frame. (1)

4.3 4.3.1 A – Cornice
B – Horizontal grounds
C – Quadrant mould (3)

4.3.2
- It gives a pleasing, decorative and durable appearance.
- It conceals cracks in the walls.
- Panelling does not require the entire wall to be plastered.
- It can be fixed directly to the walls.
- It provides insolation
- Keeps the room cool in the summer and warm in the winter.
- Expansion and shrinkage are minimal. (Any THREE) (3)

4.4
- Shutterboards
- Plywood
- Blockboard (Any TWO) (2)

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QUESTION 5: CENTRING, FORMWORK, SHORING SUSPENDED FLOORS AND IRONMONGERY (SPECIFIC)

5.1 5.1.1 A – Rib
B – Close lagging
C – Plywood or hardboard (3)

5.1.2 Nailed under the ends of the ribs to keep the ribs together. (1)

5.2 5.2.1 By lining the laggings with plywood or hardboard. (2)

5.2.2 A – Half round collar or yokes
B – Laggings
C – Indent for vertical clamps or cleats (3)

5.2.3
- Formwork refers to a temporary framework made out of timber or steel, which is ideal for the in-situ casting of concrete.
- It provides support until the concrete has set.
- Striking can be defined as the dismantling of formwork once the concrete has cured and sufficiently hardened. (4)

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5.3 5.3.1 Single flying shore (1)

5.3.2
- Raker / raking strut is inclined timber pieces.
- The raker transfers the horizontal and vertical load exerted on the wall to the sole plate. (2)

5.3.3 A – Wall
B – Wall plate
C – Needle
D – Cleat
F – Folding wedges (5)

5.3.4
- To support leaning or unstable walls and columns by transferring additional weight
- From the raking strut / raker to the ground or other supporting members, away from the wall
- Flying shores are used to provide temporary horizontal support to two parallel walls
- Where one of the walls tends to lean or show signs of failure (4)

5.4 5.4.1 Parliament hinge (1)

5.4.2 A – Leaf
B – Staggered countersunk hole for screw (2)

5.4.3 The hinges protrude toward the outside face of the walls, allowing doors to open up to 180° past the reveal or other obstructions against a wall. (2)

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QUESTION 6: SUSPENDED FLOORS, CEILINGS AND STAIRCASES (SPECIFIC)

6.1
- Span of the floor
- The centre-to-centre spacing between the floor joist
- The grade of timber of the floor joist (3)

6.2
- Floorboard: A soft timber board used as a covering and should not be more than 25 mm wide and not more than 140 mm wide.
- Floor joist: The horizontal beams that are the primary structural members in the construction of a suspended timber floor. (4)

6.3 6.3.1 Secret nailing at a 45° angle. (1)

6.3.2
- Floorboards should be secret – nailed to each floor joist with oval wire nails.
- Secret nails are driven into the edge of the floorboards at an angle of approximately 45° to the floorboard.
- The heads of these nails are punched flush with the edge of the floorboard. (4)

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6.4 6.4.1 A – Tie beam
B – Brandering
C – Ceiling board
D – Coverstrip (4)

6.4.2 To support the ceiling (1)

6.4.3 Underneath the ceiling board against the internal wall (2)

6.4.4
- Insulation ceiling board to keep heat in rooms / to provide thermal insulation.
- Fibre-cement ceiling board to waterproof ceilings for rooms / used outdoors under verandas and eaves. (4)

6.5
- Ceiling boards must always be fixed with the length of the board at right angles to the branders.
- Plasterboard is fixed with the printed side up for direct decoration or for plastering.
- Always nail or screw from the centre of the board outwards. (4)

6.6 6.6.1 F – The horizontal distance measured from the face of a rise to the face of the next riser.

6.6.2 C – The vertical post that holds up the handrail.

6.6.3 A – The inclined parts used in timber staircases to support the steps.

6.6.4 E – The distance between the top of the string and the pitch line.

6.6.5 D – A template, made out of plywood or other board products, used to set out a staircase. (5)

6.7 6.7.1 The vertical member between two consecutive treads (1)

6.7.2 Vertical distance from the top of the tread to the top of the next tread (1)

6.7.3 A flat platform at the top of a flight of stairs (1)

6.8 Timber boards used to cover the floor joist and trimmers exposed by the stairwell openings. (3)

6.9 For safety reasons. (2)
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TOTAL: 200

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ANSWER SHEET 1	CIVIL TECHNOLOGY WOODWORKING	NAME : _____

3.2 Calculate the volume of concrete needed to cast the floor slab between the external walls. (12)

A	B	C	D
			Internal measurements of long walls
			= 9 000 mm – 220 mm✓ – 220 mm✓
			= 8 560 mm ✓ **(3)**
			Internal measurements of short walls
			= 5 000 – 220 mm✓ – 220 mm✓
			= 4 560 mm ✓ **(3)**
			Volume of concrete needed
1/ ✓	**8,56 ✓**		**Length of floor slab = 8 560 mm**
	4,56 ✓		**Width of floor slab = 4 560 mm**
	0,085 ✓	**3,318 m³ ✓✓**	**Thickness of floor slab = 85 mm** **(6)**
			(12)

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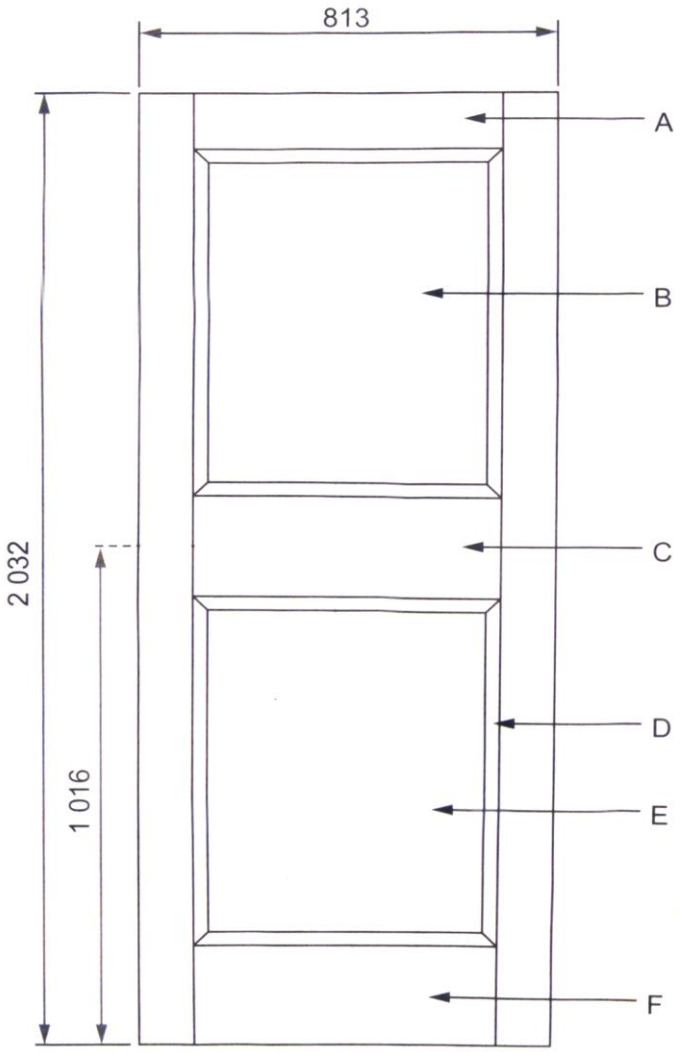
ANSWER SHEET 2	CIVIL TECHNOLOGY WOODWORKING	NAME: _____

4.1 Incomplete front elevation of a two panel door with middle / lock rail.



NOT TO SCALE

ASSESMENT CRITERIA	MARKS	CANDIDATE'S MARKS
Top rail (A)	2	
Top flat panel (B)	2	
Middle/Lock rail (C)	3	
Quadrant/Quarter-round (D)	2	
Bottom flat panel (E)	2	
Bottom rail (F)	1	
Measurements	3	
Scale	2	
Parts x 3	3	
TOTAL:	**20**	

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